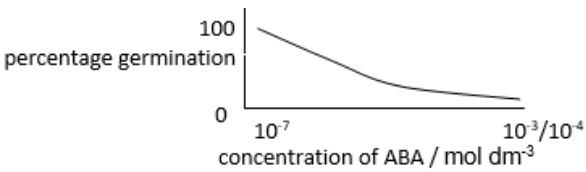


1(a)(i)	1 distilled / deionised / pure / AW, water (as diluent) ; 2 (use) 0.264g ABA and (dissolve in) 1 dm³ water ;	2
1(a)(ii)	1 add, 10 cm³ of, stock solution / 1×10^{-3} (mol dm⁻³ ABA), to 90 cm³ (distilled) water (to make 1×10^{-4} mol dm⁻³ ABA) ; 2 description, of one correct subsequent step of the serial dilution ;	2
1(b)(i)	<i>independent variable:</i> concentration of ABA ; <i>dependent variable:</i> <u>number</u> of grains germinated or length of shoot ;	2

Question	Answer	Marks												
1(b)(ii)	any five from: (use of a control for one group of 50 grains) replace the ABA solution with, distilled / deionised / pure / AW, water ; <i>ref. to</i> (placing wheat grains in a container with) soil / paper / AW ; - carry out experiment in the dark ; - method to maintain a, constant / stated, temperature (for germination) ; <i>ref. to</i> adding the, same / stated, volume / mass, of water (to container of wheat grains) ; - for each ABA concentration, count / record / AW, the <u>number</u> of (wheat) grains that have germinated, after a, same / stated, time or for each ABA concentration, measure / record / AW, the length of the shoot, after a, same / stated, time ; - safety comment with hazard and risk and precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>wheat grains</td><td>irritant / allergy</td><td>gloves / mask / PPE</td></tr> <tr> <td>ABA / GA</td><td>irritant / allergy / toxic</td><td>gloves / mask / PPE</td></tr> <tr> <td>soil</td><td>biohazard / pathogens / allergy / irritant</td><td>gloves / mask / PPE</td></tr> </tbody> </table>	hazard	risk	precaution	wheat grains	irritant / allergy	gloves / mask / PPE	ABA / GA	irritant / allergy / toxic	gloves / mask / PPE	soil	biohazard / pathogens / allergy / irritant	gloves / mask / PPE	5
hazard	risk	precaution												
wheat grains	irritant / allergy	gloves / mask / PPE												
ABA / GA	irritant / allergy / toxic	gloves / mask / PPE												
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1(c)(i)	<u>number</u> of germinated grains, divided by the, total number of grains, x 100 ;	1												

Question	Answer	Marks
1(c)(ii)	1 axes correctly orientated with labels and units ; 2 correct values on x-axis ; 3 line shows percentage germination decreases as ABA concentration increases ; 	3
1(d)(i)	any two from: 1 <i>idea of</i> smaller standard error (SE) shows that (the mean is) more reliable ; ora 2 <i>idea of</i> smaller SE shows that (calculated / sample) mean is closer to the, true / actual, mean ; ora 3 suitable example from Table 1.1 ; 4 correct <i>ref. to</i> overlap of SE from Table 1.1 ; <i>for example:</i> <i>idea of</i>, SE for 1×10^{-7} ABA concentration and 1×10^{-6} ABA concentration do not overlap, (therefore mean shoot length may be) <u>significantly different</u> or <i>idea of</i>, SE for 1×10^{-6} ABA concentration and 1×10^{-4} ABA concentration overlap, (therefore mean shoot length) is <u>not significantly different</u>	2
1(d)(ii)	there is no difference, in the (mean) shoot length (of the seedlings grown from wheat grains with), no GA (treatment) and (1×10^{-5} mol dm ⁻³) GA (treatment) ;	1

Question	Answer	Marks
1(d)(iii)	1 use 38 degrees of freedom ; 2 compare (calculated) t -value with, critical (t -) value at, (p) = 0.05 / 5% ; 3 (if calculated) t -value is higher (than the critical t -value then) and reject null hypothesis / there is a significant difference ; ora	3

Question	Answer	Marks
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